

Revision — three-dimensional shapes

The families, the counts, the nets, the volumes and the three views, in one block.

LESSON 185–188

4 × 40 MIN

MATEMATIKA 6, TAKRORLASH

S7 8 CONSOLIDATION

LESSON CLOCK

30'

35'

35'

35'

20'

5

The families and their counts	30 min
Nets and surface areas	35 min
Volumes	35 min
Views and drawings	35 min
The mistakes that cost marks	20 min
Homework	5 min

LEARNING OBJECTIVES

- Name any common solid and give its faces, edges and vertices.
- Draw or describe its net and find the surface area from it.
- Find the volume of any prism, including a cylinder.
- Move between an isometric drawing and the three views.

TEXTBOOK

National	pp. 532–541
Cambridge	Stage 7, pp. 84–96
Workbook	Review 8

01

Explanation

The families and their counts

SOLID	FACES	EDGES	VERTICES	$F + V - E$
cube or cuboid	6	12	8	2
triangular prism	5	9	6	2
hexagonal prism	8	18	12	2
square pyramid	5	8	5	2
hexagonal pyramid	7	12	7	2
prism on n sides	$n + 2$	$3n$	$2n$	2
pyramid on n sides	$n + 1$	$2n$	$n + 1$	2

THE CURVED SOLIDS STAND OUTSIDE THE PATTERN

A cylinder has 2 flat faces and 1 curved surface; a cone 1 and 1; a sphere 0 and 1. $F + V - E = 2$ applies to none of them.

Nets and surface areas

SOLID	NET	WORKED EXAMPLE	SURFACE AREA
cube	6 squares	edge 4 cm	96 cm^2
cuboid	3 pairs of rectangles	$5 \times 4 \times 3 \text{ cm}$	94 cm^2
triangular prism	2 triangles, 3 rectangles	3-4-5 ends, 10 cm long	132 cm^2
cylinder	2 circles, 1 rectangle	$r = 5, h = 10$	471 cm^2
square pyramid	1 square, 4 triangles	base 6, slant 5	96 cm^2
cone	1 circle, 1 sector	$r = 3, \text{slant } 5$	75.4 cm^2

TWO MEASUREMENTS ARE EASY TO CONFUSE

In a cylinder the rectangle's width is $2\pi r$, not r . In a pyramid the triangles use the slant height, not the height of the solid. Almost every lost mark in this block is one of those two.

Volumes

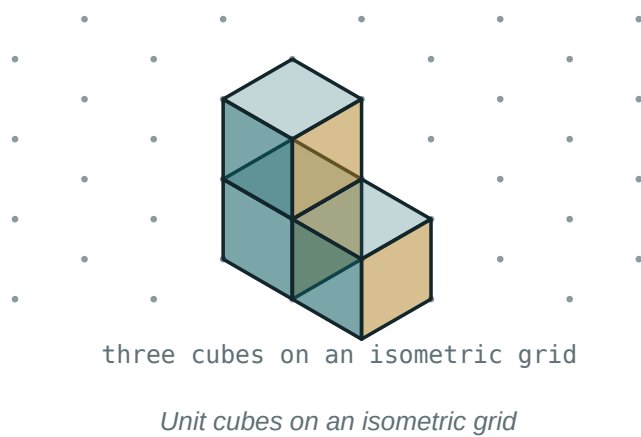
$V = \text{base area} \times \text{height}$ for every prism, including the cylinder

SOLID	BASE AREA	HEIGHT OR LENGTH	VOLUME
$5 \times 4 \times 3 \text{ cm cuboid}$	20 cm^2	3 cm	60 cm^3
cube of edge 4 cm	16 cm^2	4 cm	64 cm^3
triangular prism	12 cm^2	10 cm	120 cm^3
cylinder $r = 5$	78.5 cm^2	10 cm	785 cm^3
cylinder $r = 3$	28.3 cm^2	8 cm	226 cm^3

REVERSE IT THE SAME WAY AS BEFORE

Cross-section = $V \div \text{length}$; length = $V \div \text{cross-section}$. A prism of 240 cm^3 and length 8 cm has a cross-section of 30 cm^2 .

Views and drawings



STACK	CUBES	SURFACE	PLAN
L of three cubes	3	14	2×1
$2 \times 2 \times 1$	4	16	2×2
$2 \times 2 \times 2$	8	24	2×2
$3 \times 2 \times 2$	12	32	3×2
$3 \times 3 \times 3$	27	54	3×3

THREE VIEWS, THREE PAIRS OF DIMENSIONS

The plan gives length and depth, the front elevation length and height, the side elevation depth and height. Each dimension appears in two of the three views, which is how the drawings check one another.

The mistakes that cost marks

MISTAKE	LOOKS LIKE	CORRECT
hidden edges not counted	<i>16 edges on a hexagonal prism</i>	18
the cylinder's rectangle given width r	5×10	31.4×10
the pyramid's height used for its triangles	$4 \cdot \frac{1}{2} \cdot 6 \cdot 4$	$4 \cdot \frac{1}{2} \cdot 6 \cdot 5$
a face of the net forgotten	<i>5 faces of a cuboid</i>	6
surface area written in cm^3	$94\ cm^3$	$94\ cm^2$

THREE OF THE FIVE ARE A MISSING PIECE

An edge, a face, or the second measurement of a rectangle. Listing every part before adding is the whole defence.

02 Worked examples

EXAMPLE 1 A hexagonal prism: find its faces, edges and vertices, and check them.

1 $n = 6$, so $F = 8$, $E = 18$, $V = 12$.

2 $F + V - E = 8 + 12 - 18$.

3 $= 2$ ✓
The check catches a missed upright edge.

ANSWER

8 faces, 18 edges, 12 vertices

EXAMPLE 2 A cylinder has radius 5 cm and height 10 cm. Find its volume and its surface area, taking $\pi = 3.14$.

- 1 Base area 78.5 cm^2 , so $V = 785 \text{ cm}^3$.
- 2 Two circles 157 cm^2 , rectangle $31.4 \cdot 10 = 314 \text{ cm}^2$.
- 3 Surface 471 cm^2 .
Volume in cm^3 , area in cm^2 ✓

ANSWER

785 cm^3 and 471 cm^2

EXAMPLE 3 A prism has volume 240 cm^3 and length 8 cm. Find the area of its cross-section and describe its plan if the cross-section is a 5 by 6 rectangle.

- 1 $240 \div 8 = 30 \text{ cm}^2$.
- 2 A 5 by 6 rectangle has area 30 ✓
- 3 Standing on that face, the plan is 5 by 6.
The prism is a $5 \times 6 \times 8$ cuboid.

ANSWER

30 cm^2 ; the plan is a 5 by 6 rectangle

03 Interactive model

Work the revision from the mistake table backwards: give the wrong answers first and let the class find each error. It fixes them faster than doing the questions correctly once.

The solids block in eight questions

A hexagonal prism has how many edges?

A square pyramid has how many faces?

The volume of any prism is:

A cylinder $r = 5$, $h = 10$ holds:

157

314

471

785

Its surface area is:

157

314

471

785

A $3 \times 3 \times 3$ stack of unit cubes has surface:

27

36

54

81

The plan of a solid shows:

length and height

length and depth

depth and height

the volume

$F + V - E$ for a cone is:

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Net	Yoyilma	Развёртка
Cross-section	Kesim	Поперечное сечение
Elevation	Ko'rinish	Вид
Slant height	Yon balandlik	Апофема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A prism on an n -sided base has faces:

n

$n + 1$

$n + 2$

$2n$

A pyramid on an n -sided base has edges:

n

$n + 1$

$2n$

$3n$

The cylinder's net rectangle is as wide as:

r

$2r$

πr

$2\pi r$

The triangles of a pyramid net use:

the height

the slant height

the base

the volume

Surface area is measured in:

cm

cm^2

cm^3

litres

A count giving $F + V - E = 3$ means:

an unusual solid

a wrong count

a curved surface

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Faces, edges and vertices of a cube

ANSWER 6, 12, 8

2. Faces, edges and vertices of a triangular prism

ANSWER 5, 9, 6

3. $F + V - E$ for any flat-faced solid

ANSWER 2

4. The net of a cylinder

ANSWER Two circles and a rectangle

5. The volume of a prism

ANSWER Base area $\times$ height

6. The plan is the view from

ANSWER Above

7. The number of different nets of a cube

ANSWER 11

Medium · the standard question

7 PROBLEMS

1. Faces, edges and vertices of a hexagonal prism

ANSWER 8, 18, 12

2. Faces, edges and vertices of a square pyramid

ANSWER 5, 8, 5

3. The volume of a $5 \times 4 \times 3$ cm cuboid

ANSWER 60 cm^3

4. Its surface area

ANSWER 94 cm^2

5. A prism of cross-section 12 cm^2 , length 10 cm

ANSWER 120 cm^3

6. A cylinder $r = 5$, $h = 10$, $\pi = 3.14$

ANSWER 785 cm^3

7. The surface of a $2 \times 2 \times 2$ block of unit cubes

ANSWER 24

Hard · stretch and reason

7 PROBLEMS

1. The surface area of a cylinder $r = 5$, $h = 10$

ANSWER 471 cm^2

2. The surface area of a 3-4-5 triangular prism 10 cm long

ANSWER 132 cm^2

3. The surface area of a square pyramid, base 6, slant 5

ANSWER 96 cm^2

4. Cubes in a $3 \times 3 \times 3$ stack, and how many are hidden

ANSWER 27 and 1

5. A prism of volume 240 cm^3 and length 8 cm: its cross-section

ANSWER 30 cm^2

6. The edges of a prism on an n -sided base

ANSWER $3n$

7. Why can three views not fix the number of cubes?

ANSWER A hidden cube changes none of them

07 Homework

Homework — 5 tasks

List every face of a net before adding, and check every set of counts with $F + V - E = 2$.

1. Give the faces, edges and vertices of a pentagonal prism and of a pentagonal pyramid.
2. Find the surface area of a cuboid 7 cm by 5 cm by 3 cm from its net.
3. A cylinder has radius 3 cm and height 10 cm. Find its volume and surface area.
4. A triangular prism has a cross-section of area 18 cm^2 and is 9 cm long. Find its volume.
5. Draw a $3 \times 2 \times 1$ stack of cubes and give its plan and two elevations.